main.c

Run ▶

```c
#include <stdio.h>
#include <string.h>

int main() {
    char nonTerminal;
    char alpha[50], beta[50];

    printf("Enter the non-terminal: ");
    scanf(" %c", &nonTerminal);

    printf("Enter alpha (the part after left recursion): ");
    scanf("%49s", alpha);

    printf("Enter beta (the non-left-recursive part): ");
    scanf("%49s", beta);

    printf("\nGrammar before eliminating left recursion:\n");
    printf("%c -> %c%s | %s\n", nonTerminal, nonTerminal, alpha,
beta);

    printf("\nGrammar after eliminating left recursion:\n");
    printf("%c -> %s%c'\n", nonTerminal, beta, nonTerminal);
    printf("%c' -> %s%c' | epsilon\n", nonTerminal, alpha,
nonTerminal);

```

Output

```
Enter the non-terminal: A
Enter alpha (the part after left recursion): a
Enter beta (the non-left-recursive part): b

Grammar before eliminating left recursion:
A -> Aa | b

Grammar after eliminating left recursion:
A -> bA'
A' -> aA' | epsilon
```

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Run ▶

```c
int main() {
    char nonTerminal;
    char alpha[50], beta[50];

    printf("Enter the non-terminal: ");
    scanf(" %c", &nonTerminal);

    printf("Enter alpha (the part after left recursion): ");
    scanf("%49s", alpha);

    printf("Enter beta (the non-left-recursive part): ");
    scanf("%49s", beta);

    printf("\nGrammar before eliminating left recursion:\n");
    printf("%c -> %c%s | %s\n", nonTerminal, nonTerminal, alpha, beta);

    printf("\nGrammar after eliminating left recursion:\n");
    printf("%c -> %s%c'\n", nonTerminal, beta, nonTerminal);
    printf("%c' -> %s%c' | epsilon\n", nonTerminal, alpha, nonTerminal);

    return 0;
}
```

Output

```
Enter the non-terminal: A
Enter alpha (the part after left recursion): a
Enter beta (the non-left-recursive part): b

Grammar before eliminating left recursion:
A -> Aa | b

Grammar after eliminating left recursion:
A -> bA'
A' -> aA' | epsilon
```